

As discussed before (Sec. 24.2 in the discussion concerning compensating the op-amp), when we get a decrease in the 3-dB frequency and an increase in the low-frequency gain, the effects cancel and the gain-bandwidth product ( $f_{un}$ ) remains constant. The unity-gain frequency is still given by Eq. (24.44). For the OTA in Fig. 24.35, we can estimate the low-frequency gain, using Tables 9.2 and 20.1, as

$$A_v = (150) \cdot (16.6||4.2) = 500 \text{ (54 dB)}$$

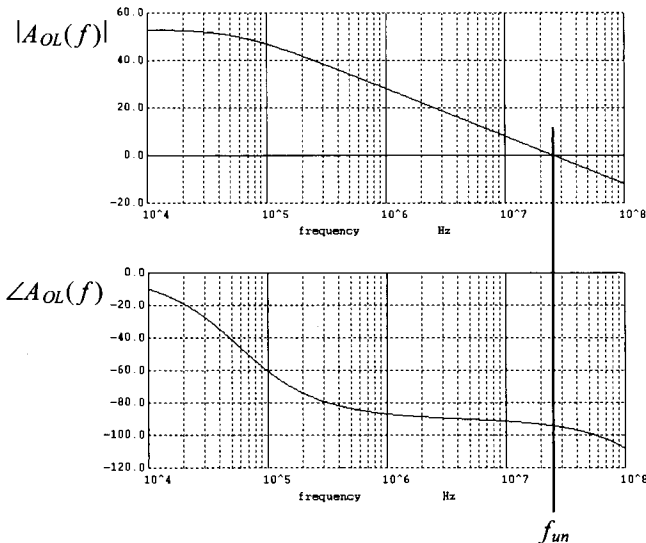
and the 3-dB frequency, driving a 1 pF load, is

$$f_{3dB} = \frac{1}{2\pi(16.6||4.2 \times 10^6)(1 \text{ pF})} = 47 \text{ kHz}$$

Simulation results are seen in Fig. 24.36. The PM is roughly 85°.

### An Important Note

Note, in Eq. (24.44), that increasing the load capacitance, decreases the unity-gain frequency, making the OTA more stable. Unlike the two-stage op-amps discussed in the first two sections of this chapter, where the op-amp can become unstable with a large load capacitance, the OTA only becomes more stable with large load capacitance. Also, the PM approaches 90° as the load capacitance increases. The step response of the OTA has a first-order shape (just as in an RC circuit). In many on-chip applications, the load is purely capacitive and so the OTA works great as an “op-amp” in a closed-loop configuration (thus we’ll often call OTAs op-amps). Note that adding a second stage to the basic OTA forms a (two-stage) op-amp like those discussed earlier. We can use the methods already presented to compensate the resulting structure (but, again, driving a load capacitance that’s too large with these two-stage structures can result in instability).



**Figure 24.36** The open-loop gain and phase response for the OTA in Fig. 24.35.

### OTA with an Output Buffer (An Op-Amp)

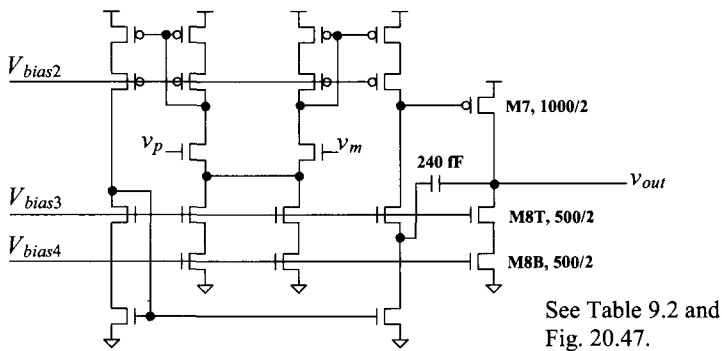
If we use the OTA in Fig. 24.35 (or Fig. 24.33) to drive a 1 pF load, we're limited to charging/discharging the capacitor at a rate of

$$\frac{I_{SS}}{C_L} = \frac{dv_{out}}{dt} = \frac{20 \mu A}{1 pF} = \frac{20 mV}{ns} \quad (24.48)$$

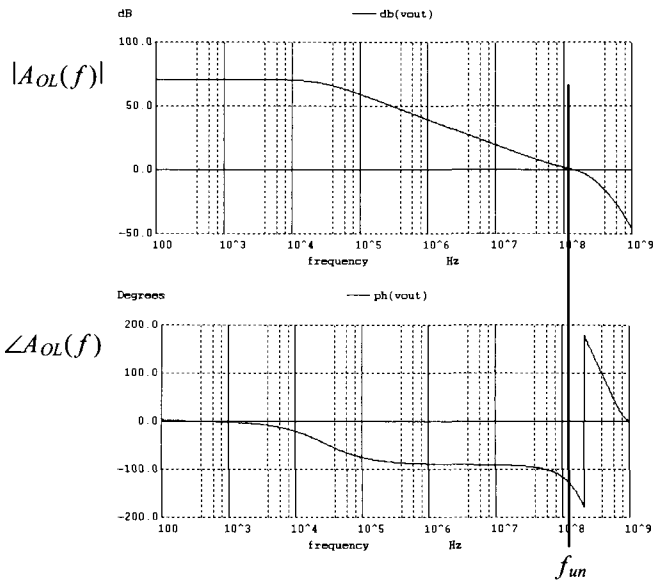
For the OTA's output to change from ground to  $V_{DD}$  (1 V) requires (just due to slewing) 50 ns. This length of time (the slew-rate limited portion of the settling time) may be fine for many applications. However, for high-speed design, we want faster settling times. Looking at Eq. (24.48), we see that we can decrease the settling time by increasing  $I_{SS}$ . However, as discussed in Ch. 9, this causes the gain of the circuit to decrease. To avoid the reduction in gain, we can make a corresponding increase in the size of the devices to keep the overdrive voltage constant. But this is doing nothing more than effectively reducing the size of the load capacitor (the load capacitance becomes comparable in size to the parasitic transistor capacitances).

To speed up the circuit, we may add an output stage, Fig. 24.37. The added stage is a common-source amplifier (with a negative gain indicating **we swap the inverting and noninverting terminals** of the op-amp or else we waste a lot of time). While M7 can turn on and charge a load capacitor quickly, the current source, M8, still limits the rate of load capacitance discharge. For this reason, we've bumped up the current in the output stage to 100  $\mu A$ . The open-loop simulations driving a 1-pF load are seen in Fig. 24.38. The PM is  $45^\circ$  (too low if the op-amp will be used in a unity follower configuration). Figure 24.39 shows the step response of the op-amp.

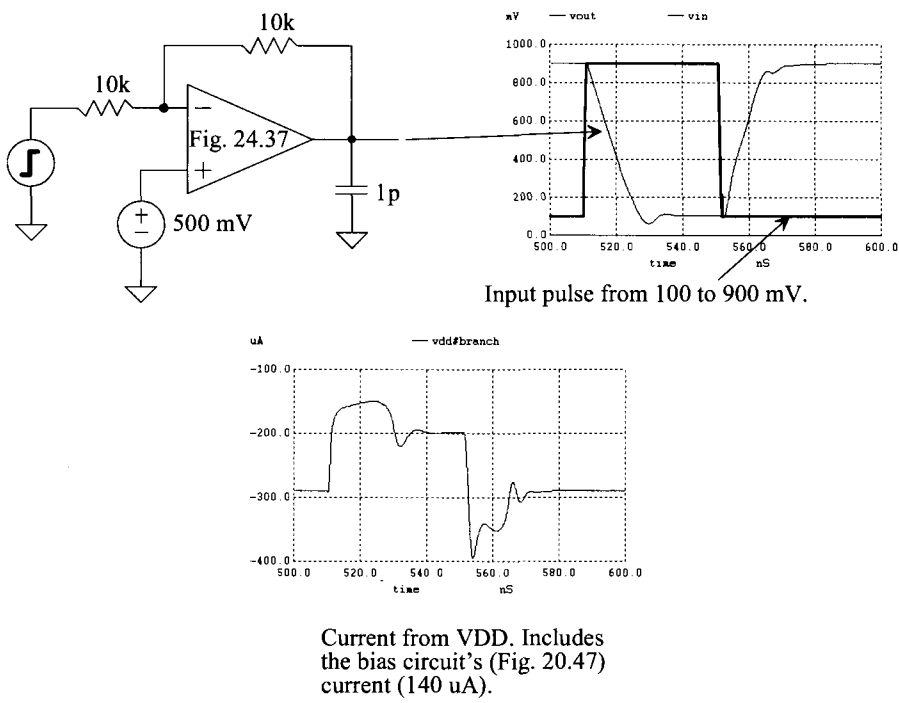
We should compare this figure to the response we saw in Fig. 24.31. The op-amp in Fig. 24.31 is driving a heavier load but still has cleaner settling behavior. Further, a comparison of the current pulled from  $V_{DD}$  reveals, under the same loading and operating conditions, that the op-amp in Fig. 24.37 pulls more current (see Fig. 24.39). The main difference in these two op-amps is the output buffer used. Let's consider adding a class AB buffer to the OTA in Fig. 24.35 (like the one used in the op-amp of Fig. 24.29). Since we are only driving a 1 pF load, we'll leave the output devices, MON and



**Figure 24.37** A cascode OTA circuit with common-source output buffer (an op-amp). Note the inverting and noninverting terminals are swapped from Fig. 24.35.



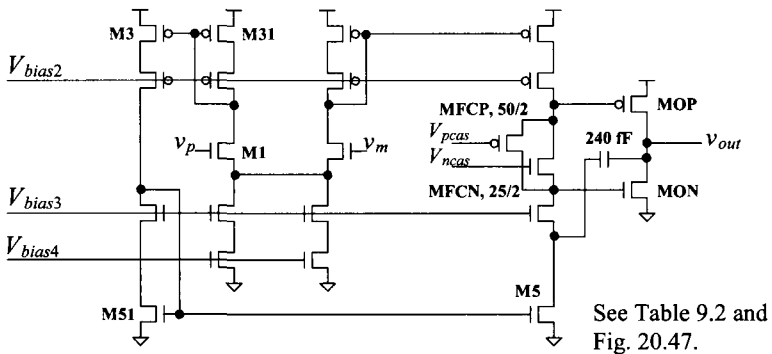
**Figure 24.38** Open-loop response of the op-amp in Fig. 24.37 driving a 1-pF load.



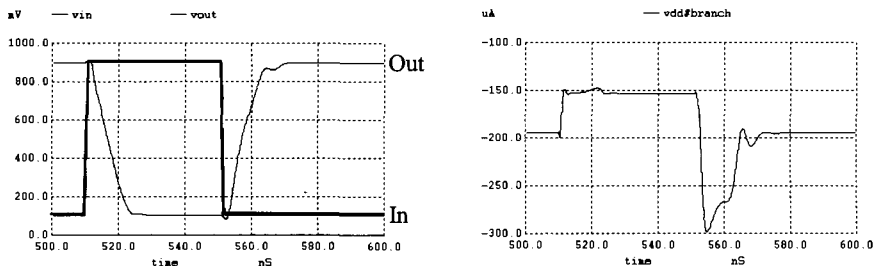
**Figure 24.39** Step response of the op-amp in Fig. 24.37 driving a 1 pF.

MOP, at the same size as the other MOSFETs in the circuit (PMOS are 100/2 and NMOS are 50/2). The resulting op-amp is seen in Fig. 24.40. The step response (in the same configuration used to generate Fig. 24.39) is seen in Fig. 24.41. Notice that the current pulled from  $V_{DD}$  by the op-amp in Fig. 24.40 is 100  $\mu\text{A}$  less than the current pulled by the op-amp in Fig. 24.37. This lower current is the result of using a class AB output stage.

The performance of the op-amps in Figs. 24.37 and 24.40, with regard to high-speed operation and settling time, won't be quite as good as the topology in Fig. 24.29 (assuming the same biasing conditions, device sizes, and overdrive voltages). To understand why, look at the path, in Fig. 24.40, that the noninverting op-amp input,  $v_p$ , takes to get to the op-amp output,  $v_{out}$  (that is, M1 - M31 - M3 - M51 - M5 - MON). The delay through this path is longer than the delay, in Fig. 24.29, through M2 - MOP. Further, for high speed, M5 of the op-amp in Fig. 24.40 must turn on (above its quiescent current level) to pull the gate of MON down. For the op-amp in Fig. 24.29, the fact that both the gate of M9T and MOP are driven (in opposite directions) by the diff-amp allows us to use a simple pull-down current (M8) without losing speed. The issues with the cascode diff-amp-based, op-amp topology of Fig. 24.29 are the more limited input common-mode range and the limited voltage swing of node 1. Towards getting better input common-mode voltage range and output swing as well as improved high-speed operation, let's discuss the folded-cascode OTA (see Fig. 22.5).



**Figure 24.40** Using a class AB output stage with the OTA.

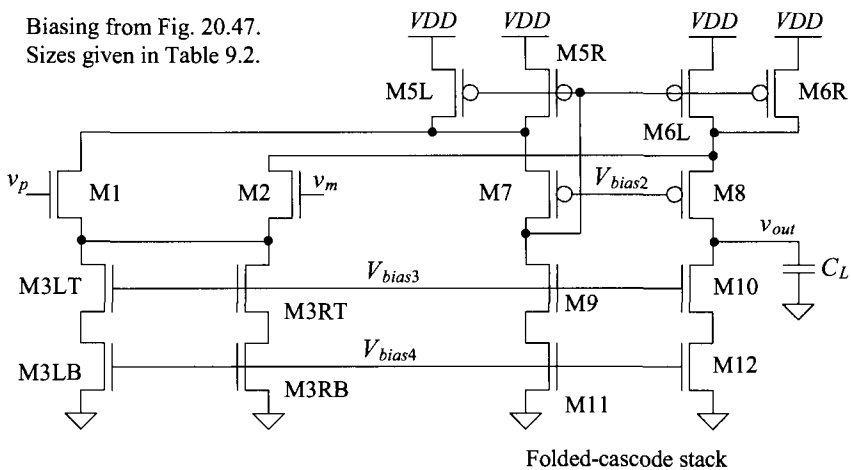


VDD current (includes 140  $\mu\text{A}$  from bias circuit).

**Figure 24.41** The step response and current in the topologies seen in Fig. 24.39 using the op-amp in Fig. 24.40.

### The Folded-Cascode OTA and Op-Amp

An example of an NMOS diff-amp-based folded cascode OTA is seen in Fig. 24.42. It is assumed that the reader understands the biasing and operation of the circuits seen in Figs. 20.45, 20.46, and 22.5. To avoid labeling MOSFETs with twice the width, we've drawn the schematic in Fig. 24.42 with MOSFETs in parallel in the branches that supply or sink more current (for example, the diff-amp's tail current). All MOSFETs in Fig. 24.42 conduct 10  $\mu\text{A}$  of current under equilibrium conditions. Before moving on, it may be a good idea to pause and make sure that the reader understands how the currents sum in this OTA.



**Figure 24.42** A folded-cascode OTA.

To qualitatively describe the operation of the OTA, let's look at what happens when  $v_p$  goes high (above  $v_m$ ). This causes M1 to turn on and M2 to shut off. When M1 turns on, it pulls the drain of M5 down, shutting M7 off. As M7 shuts off, M9 and M11 pull the gate of M5 down. With the gate of M5 being pulled down, the current in M6 increases. At the same time, because the current in M2 is decreasing, the current in M8 is increasing and the output voltage increases. The maximum current the OTA output can supply (out of the OTA to the load) is in the neighborhood of 20  $\mu\text{A}$ , while the maximum current the OTA can sink (into the OTA) is 10  $\mu\text{A}$  (set by the current sink M10/M11).

The same AC equations presented at the beginning of this section also apply to this OTA. For example, the AC drain currents of M1 and M2 can be written as

$$i_{d1} = -i_{d2} = g_{mn}v_{gs1} = -g_{mn}v_{gs2} \quad (24.49)$$

which, again, indicates  $v_{gs1} = -v_{gs2}$ . However, we know

$$v_p - v_m = v_{gs1} + (-v_{gs2}) \quad (24.50)$$

so

$$i_{d1} = i_{d5} = i_{d6} = (v_p - v_m) \cdot \frac{g_m}{2} \quad (24.51)$$

noting that the AC current through M9/M11 (and thus M7) is ideally zero. The current through M8 is then

$$i_{d8} = i_{d6} - i_{d2} = 2 \cdot i_{d1} \quad (24.52)$$

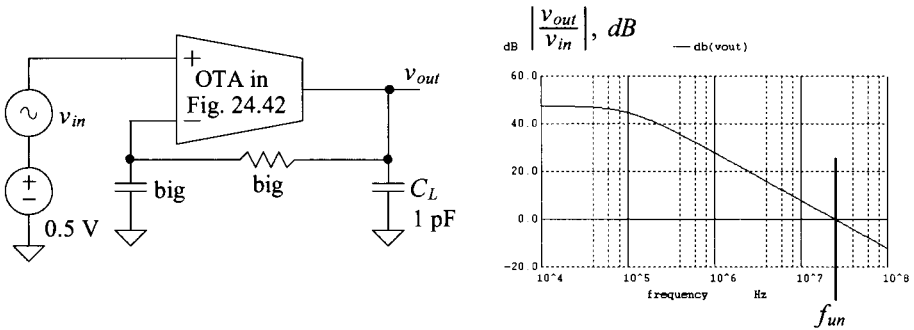
and thus the output voltage (again, the AC current in M10/M11 is ideally zero) is

$$v_{out} = i_{d8} \cdot (R_{ocasn} || R_{ocasp}) \quad (24.53)$$

and thus the gain is

$$A_v = \frac{v_{out}}{v_p - v_m} = g_{mn} \cdot (R_{ocasn} || R_{ocasp}) \quad (24.54)$$

This result should be compared to Eq. (24.46). Equation (24.47) gives the 3-dB frequency of the OTA's open-loop response, while Eq. (24.44) gives the unity-gain frequency. Figure 24.43 shows the open-loop response of the folded-cascode OTA driving a 1 pF load capacitance.



**Figure 24.43** Simulating the open-loop response of the OTA in Fig. 24.42.

Figure 24.44 shows the schematic of a folded-cascode op-amp using a class AB output buffer. We've added floating current sources to equalize the voltages across M9/M10 (to minimize the input-referred offset). In the open-loop response of the op-amp (see Fig. 24.45), the load of the op-amp is a 1 pF capacitor. The step response is seen in Fig. 24.46, using the topology of Fig. 24.39 (gain of  $-1$  driving 1 pF and the 10k feedback resistor). Reviewing the previous op-amp responses, we don't see much difference in performance. We can't seem to push the gain-bandwidth product,  $f_{un}$ , much above 100 MHz. Again, the only way to increase the speed is to reduce the channel length and/or increase the overdrive voltage.

Figure 24.47 shows what happens if we reduce the lengths of the MOSFETs used in the op-amp in Fig. 24.44 from 2 (100 nm) to 1 (50 nm). The gain-bandwidth product jumps to 400 MHz. However, the low-frequency gain drops to approximately 50 dB. The step response is clean, and the settling time is approximately 5 ns. While we can increase the speed further by redesigning the bias circuit and increasing the currents and overdrive voltage, let's, instead, concentrate on increasing the low-frequency gain of the op-amp. This won't increase the gain-bandwidth product of the op-amp, but it's necessary for precision amplification. To *enhance the gain*, we'll try to increase the OTA output resistance ( $R_1$  or the resistance on node 1) by regulating the drain in the cascode current sources. We've already discussed this technique (see Fig. 20.40), but we haven't applied it to amplifiers. Before leaving this section, let's present one more example.

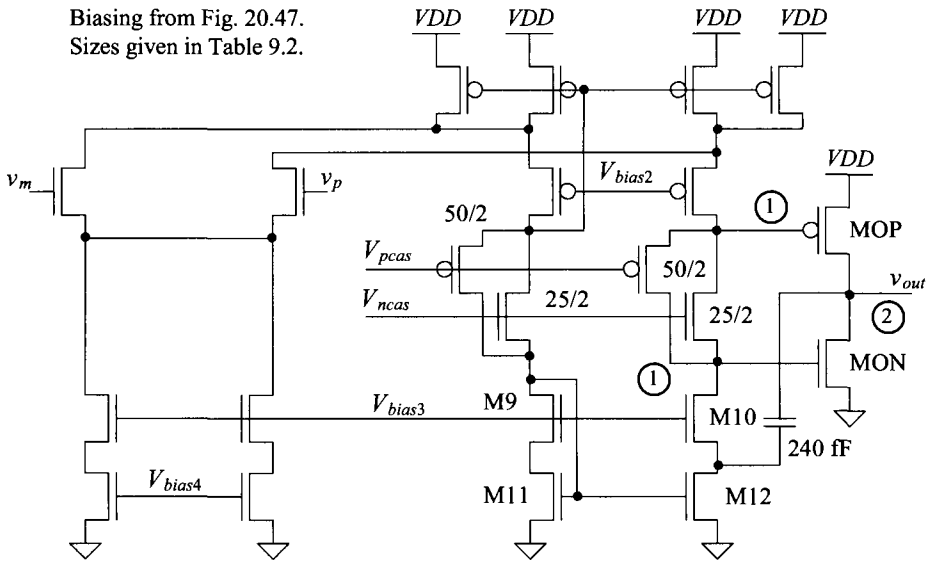


Figure 24.44 Folded-cascode op-amp with class AB output buffer.

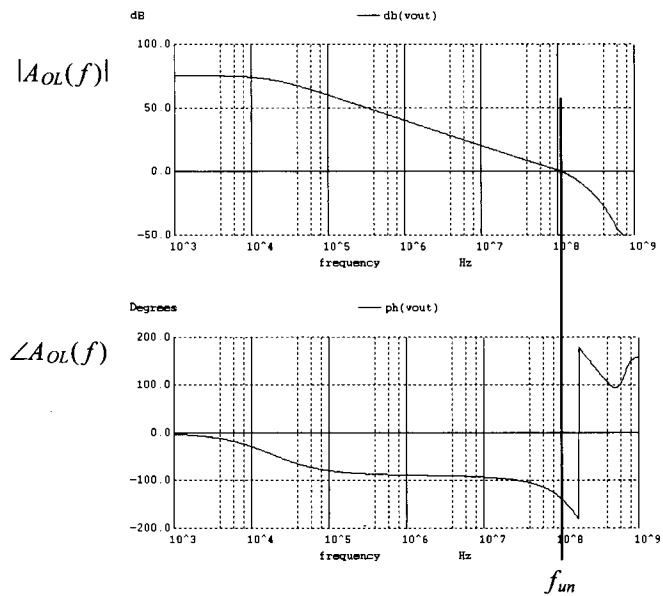
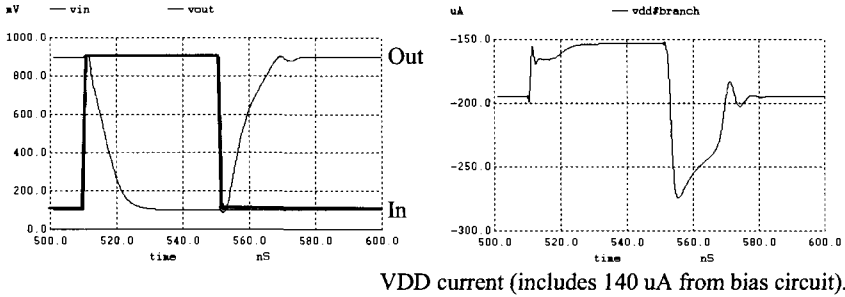
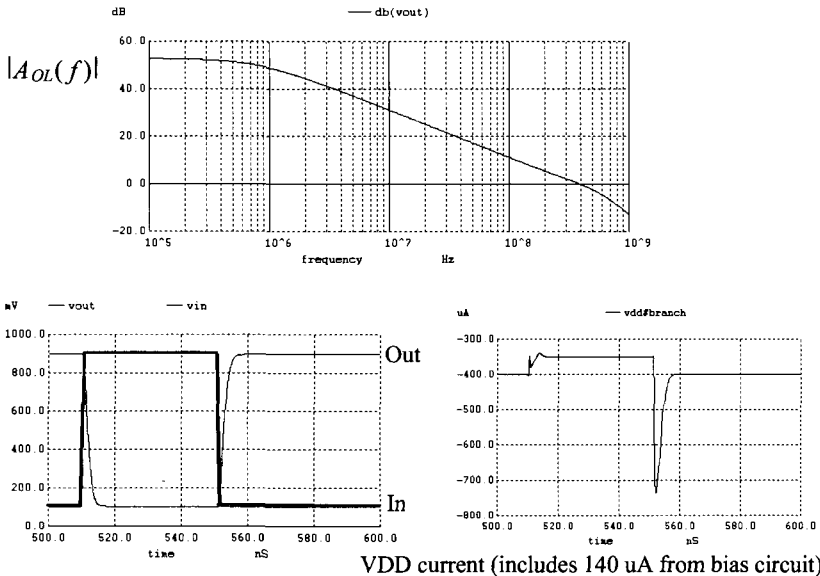


Figure 24.45 Open-loop response of the op-amp in Fig. 24.44 driving a 1-pF load.

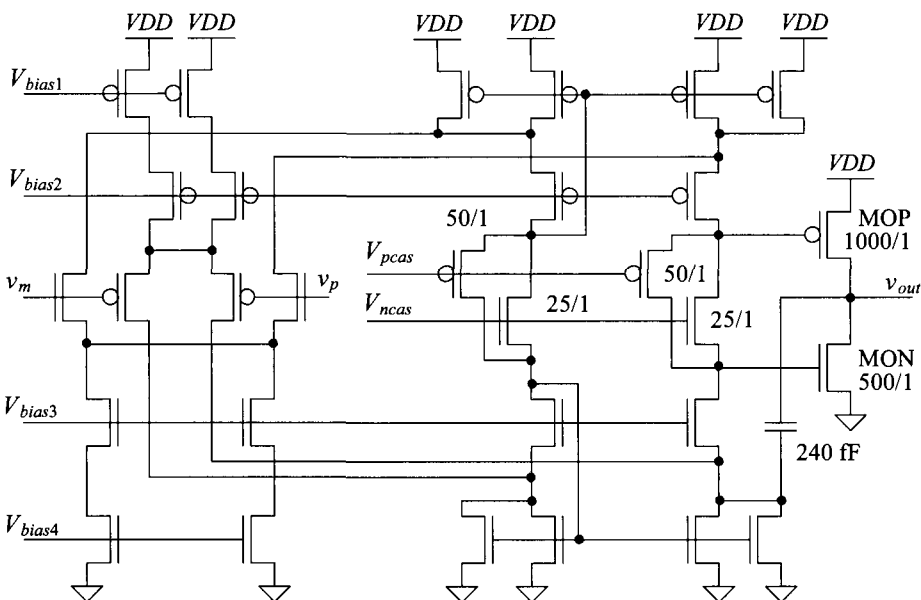


**Figure 24.46** The step response and current in the topologies seen in Fig. 24.39 using the op-amp in Fig. 24.44.

Figure 24.48 shows a wide-swing op-amp based on the topology seen in Fig. 24.44. To get wide-swing operation (which means that the input common-mode voltage extends beyond the power supply rails and the op-amp still functions), we added a PMOS diff-amp stage. Further, to sink the additional current supplied by the PMOS diff-amp when it's on, we add two NMOS devices at the bottom of the folded-cascode stack. When, for example, the PMOS diff-amp shuts off with the input common-mode voltage moving towards  $V_{DD}$ , the added two devices don't really affect the circuit's operation. The only result is that the  $V_{GS}$  of the bottom NMOS devices is slightly smaller when the PMOS diff-amp shuts off. A similar conclusion can be drawn when the NMOS diff-amp shuts off with the common-mode voltage moving towards ground. Note that when both diff-amps are on, the first-stage transconductance becomes,  $g_{mn} + g_{mp}$ . This means the gain-bandwidth product,  $f_{un}$ , moves to  $(g_{mn} + g_{mp})/2\pi C_c$ . The value of the compensation



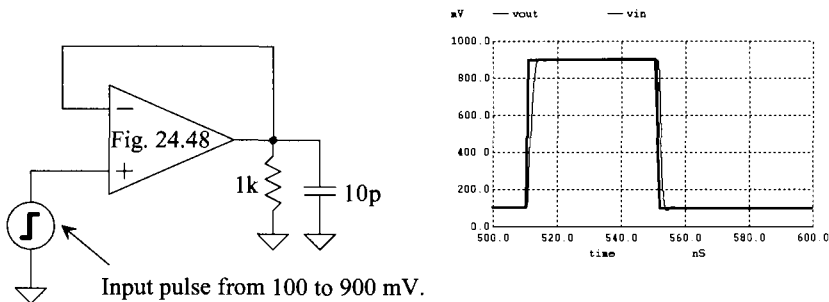
**Figure 24.47** Reducing the length of the MOSFETs used in the op-amp in Fig. 24.44 from 2 to 1 and resimulating open-loop and step responses.



Biasing from Fig. 20.47.  
 Unlabeled NMOS are 50/1.  
 Unlabeled PMOS are 100/1.

**Figure 24.48** An op-amp with an input common-mode range that extends beyond the power supply rails and that can drive heavy loads.

capacitor may need to be increased. When only one diff-amp is on, the transconductance of the other diff-amp goes to zero, resulting in a smaller gain-bandwidth product. The issue here is that the gain of the op-amp is changing with variations in the input common-mode voltage. This, as mentioned before, can lead to distortion and is the reason that the inverting topology, as seen in Fig. 24.39, is preferred (noting that the input common-mode voltage is fixed when using the inverting topology).



**Figure 24.49** Step response of the op-amp in Fig. 24.48 driving 1k and 10 pF. Note that the op-amp is in the follower configuration and that the input common-mode voltage is changing.

Finally, in some applications, an op-amp that can supply significant amounts of current, say 100 mA, may be needed. To get such a large current, we may need to increase the widths of the output MOSFETs in the push-pull amplifier to, for example, 10,000 and 5,000 (for the PMOS and NMOS devices, respectively). The problem with this is that the quiescent current that flows in these devices, using the op-amp configuration in Fig. 24.44 as an example, is 1 mA. To reduce this current, we can increase the lengths of the floating current sources used in the folded-cascode section. This increases their gate-source voltage drops and moves the gates of the push-pull MOSFETs towards the power supply rails (shutting them off). Care must be exercised when doing this because shutting off the output MOSFETs can result in moving  $g_{m2}$  towards zero, causing, as indicated in Eq. (24.24),  $f_2$  to move down in frequency, thus making the op-amp unstable.

## 24.4 Gain-Enhancement

While using minimum channel lengths ( $L = 1$ ) results in the fastest op-amps, the open-loop gain,  $A_{OLD C}$ , is considerably reduced. Using gain-enhancement (GE) techniques (regulating the drain node in a cascode structure, see Fig. 20.40), we boost the low-frequency gain by increasing the cascode output resistance. However, the gain-bandwidth product,  $f_{un}$ , is still  $g_m/2\pi C_c$ . (We get the speed by using a smaller compensation capacitor and/or a larger diff-amp transconductance.) Note, in Fig. 24.47 for example, that our gain-bandwidth product went up by the increase in  $g_m$  with the reduction in the channel length. Reviewing Eqs. (24.30)–(24.34), we see that if  $A_{OLD C}$  goes up,  $f_{3dB}$  goes down (and  $f_{un}$  is a constant).

In the following discussion, we only concern ourselves with two-stage op-amps in which we are using GE to compensate for the low gain caused by our minimum channel lengths. While GE can be used in an OTA, the fact that the speed of the OTA is limited by the capacitance it is driving keeps them from being as fast as the two-stage op-amps we've discussed (unless, of course, the load capacitance is small).

As seen in Fig. 20.40, we need an additional amplifier to help regulate the drain of the cascode device. Consider the two differential amplifiers (OTAs) seen in Fig. 24.50. The source-followers on the inputs of the diff-amps are used to allow for the amplification of signals near ground (for the P amplifier) or  $V_{DD}$  (for the N amplifier). The gains of the diff-amps are

$$A_P = g_{mp} \cdot (r_{on} || r_{op}) \text{ and } A_N = g_{mn} \cdot (r_{on} || r_{op}) \quad (24.55)$$

To simplify the equations, we'll assume

$$A_{GE} = A_P = A_N \quad (24.56)$$

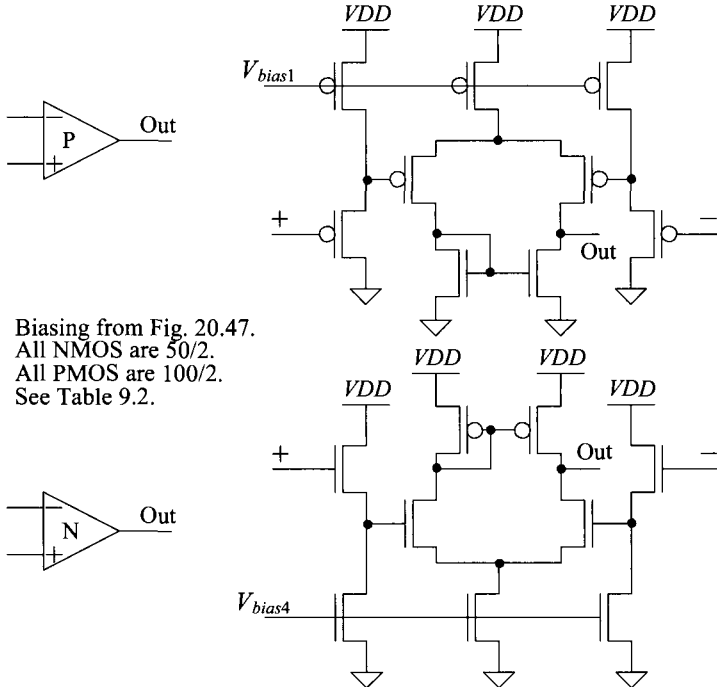
The low-frequency gain of the op-amp in Fig. 24.44 is

$$A_{OLD C} = \overbrace{g_{mn} \cdot (R_{ocasn} || R_{ocasp})}^{\text{diff-amp gain}} \cdot \overbrace{(g_{mp} + g_{mn}) \cdot (r_{on} || r_{op})}^{\text{push-pull output stage}} \quad (24.57)$$

Using Eq. (20.67), we can write the open-loop gain *with gain-enhancement* as

$$A_{OLD C, GE} = A_{OLD C} \cdot A_{GE} \quad (24.58)$$

The open-loop gain is increased by the gain of the added amplifier. Figure 24.51 shows the op-amp of Fig. 24.44 with GE.



**Figure 24.50** Diff-amps with source-follower level shifters for use in GE.

The simulation results in Fig. 24.47 were generated using the op-amp of Fig. 24.44 with the lengths reduced from 2 to 1. The low-frequency, open-loop gain was 53 dB. In Fig. 24.51 we added GE and used the resulting op-amp to generate the simulation results seen in Fig. 24.52 (with the GE compensation capacitors,  $C_{cGE}$ , set to zero). The low-frequency, open-loop gain is now 74 dB. The step response is seen as well. The current pulled from  $VDD$  is now 60  $\mu A$  higher than the op-amp without GE. By replacing the diff-amps in the N and P amplifiers in Fig. 24.50 with folded-cascode OTAs (like the one seen in Fig. 24.42), we can get a greater increase in open loop gain. However, the stability of the regulated-drain feedback loop becomes even more important.

#### *Bandwidth of the Added GE Amplifiers*

Let's model the frequency response of the added amplifiers in Fig. 24.50 by

$$A_{GE}(f) = \frac{A_{GEDC}}{1 + j \frac{f}{f_{3dBGE}}} \quad (24.59)$$

Using this equation with Eqs. (24.30) and (24.58), we can write

$$A_{OLGE}(f) = \frac{A_{OLDLC}}{1 + j \frac{f}{f_{3dB}}} \cdot \frac{\overbrace{A_{GEDC}}^{A_{GE}(f)}}{1 + j \frac{f}{f_{3dBGE}}} \quad (24.60)$$